

4(a)	1 diploid ; 2 allele ; 3 dominant (allele) ;	3
4(b)	any four from: 1 CDV acts as a selection pressure ; 2 DNA / allele, for black (coat) gives <u>resistance</u> to CDV or DNA / allele, for black (coat), gives (selective) advantage / is selected for, if / when, CDV is common / dogs are common ; <i>in the south (context must be established at least once):</i> 3 more CDV (infection) in <u>wolves</u> ; 4 black wolves, more likely to survive / have selective advantage / are selected <u>for</u> ; ORA greys 5 black wolves more likely to, breed / reproduce ; ORA greys 6 pass on / increase the frequency of, <u>allele</u> for black coat ; ORA grey	4
4(c)	1 positive (linear) correlation / positive relationship / direct relationship / directly proportional or as % with (anti-CDV) antibodies increases the % of black wolves increases ; <i>plus any three from:</i> 2 wolves that survive CDV will, later / as a result, have antibodies ; 3 high % of antibodies show population with high level of CDV ; 4 in population with high % antibodies, grey wolves have died / more black wolves are left ; 5 gene / DNA / allele / protein, that gives black coat, helps survival / gives resistance to CDV / defends against CDV ; 6 AVP ;	4